

Regional Quantitative Problem-Solving Circle

Problem Set 5: Algorithmic Invariants & Exchange Arguments (Weeks 9–10)

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Instructions: Do not rely on heuristic guessing. Every problem requires structural justification via formal Exchange Arguments or monotonic potential functions. Ensure all state transitions and partial sums are rigorously evaluated.

Problem 1 (Interval Scheduling / Adjacent Exchange — IOI / Kleinberg-Tardos Caliber). Given a set of n resource requests, where each request i has a start time s_i and a finish time f_i , a schedule is valid if no two selected requests overlap in time. Prove that the greedy algorithm—which iteratively selects the compatible request with the earliest finish time—is globally optimal and maximizes the total number of scheduled requests.

(Hint: Define a hypothetical optimal schedule O , and execute a mathematical induction over the interval index r to prove that the greedy schedule G finishes no later than O at every step. Conclude by deriving a contradiction on the cardinality $|G| = |O|$.)

Problem 2 (Minimizing Maximum Lateness / Inversion Invariant — Algorithmic Tripes). A single processor must execute a set of n independent jobs. Each job i requires a continuous processing time t_i and has a strict deadline d_i . The lateness of job i is defined as $L_i = \max(0, C_i - d_i)$, where C_i is its completion time. The goal is to minimize the maximum lateness across all jobs, $L = \max_i L_i$. Prove that the Earliest Deadline First (EDF) scheduling rule (sorting jobs such that $d_1 \leq d_2 \leq \dots \leq d_n$) is globally optimal.

(Hint: Assume an optimal schedule contains an inversion—an adjacent pair where $d_i > d_{i+1}$. Prove that swapping this inverted adjacent pair never increases the global maximum lateness.)

Problem 3 (Chip-Firing / Monovariant Termination — Olympiad Caliber). Consider a finite line graph with vertices labeled $\{1, 2, \dots, M\}$. Exactly N chips are distributed across these vertices. If any vertex v (where $1 < v < M$) contains at least 2 chips, a legal "firing" move consists of removing 2 chips from v and sending exactly one chip to $v - 1$ and exactly one chip to $v + 1$. Define the potential state function $\Phi = \sum_{v=1}^M v^2 \cdot c(v)$, where $c(v)$ represents the number of chips currently at vertex v . Using this potential function, prove that the chip-firing game must strictly terminate in a finite number of steps, regardless of the sequence of legal moves chosen.